Maghfira Ramadhani

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Summary

Applied microeconomist studying how climate change, environmental risks, and the energy transition affect firm and individual behavior. Combines economic theory, causal inference, and large-scale data to examine the roles of technology, policy, and behavioral adaptation in responding to these challenges. Brings five years of advisory experience serving government agencies, regulators, and energy companies.

Education

Georgia Institute of Technology, Ph.D. in Economics — Expected May 2027
Fields: energy and environmental economics, applied microeconomics
Advisors: Dylan Brewer and Bobby Harris (co-chairs), Casey Wichman, Laura Taylor, Dan Matisoff
Georgia Institute of Technology, M.S. in Economics — 2025
Institut Teknologi Bandung, B.S. and M.S. in Petroleum Engineering — 2021

Selected Dissertation Research

Battery Storage and Natural Gas Generator Market Power (Job Market Paper)

- Data: Generator-hour panel (2018–2024) linking ERCOT bids and dispatch, EIA unit records, EPA emissions, and fuel prices.
- Methods: Strategic pricing model; staggered difference-in-differences; instrumental variable; competitive-price profit decomposition.
- Findings: An average storage entry lowers nearby gas generators' evening markups by 4% and daily operating profits by 5%; the resulting $47 million annual transfer flows mainly to consumers.

Digital Dispatch and Demand Response in Grid Emergencies (with Soren Anderson and Dylan Brewer)

- Data: Ecobee smart-thermostat records from five western states, linked to CAISO alerts and weather during the 2022 heatwave.
- Methods: Difference-in-differences; event studies; model-based welfare analysis.
- Findings: A phone alert tripled automated demand response efficacy; the full Flex Alert episode yielded an estimated $69.8 million in net welfare gains, primarily from avoided blackouts.

Air Pollution and Behavior: Evidence from Wearable Devices (with Bobby Harris)

- Data: NIH *All of Us* Fitbit records (51 million activity participant-days), linked to health records and EPA pollution data.
- Methods: Wind-direction instrumental variable; distributed-lag models.
- Findings: $PM_{2.5}$ reduces physical activity, running performance, and sleep; subsequent recovery highlights the importance of measuring effects beyond the exposure day.

Work Experience

Graduate Researcher and Instructor — 2022–present
Georgia Institute of Technology, School of Economics, Atlanta, GA

- Led a team of Ph.D. and undergraduate research assistants on an interdisciplinary survey of renewable integration, coauthored and published in *Renewable and Sustainable Energy Reviews* (2026).
- As Research Assistant to Matthew E. Oliver (2023–2024), supported projects published in the *Journal of the Association of Environmental and Resource Economists* (2025) and book chapter in "The Economics of Integrating Distributed Energy Resources into the Electricity System" (2025).
- Instructor, Statistics for Economists (evaluation: 4.32/5); redesigned the course to develop data analysis skills [Course Website]; head teaching assistant for ten courses; Graduate Teaching Fellow at the Center for Teaching and Learning.
- Student Affiliate, Energy Policy and Innovation Center (EPICenter), Georgia Tech (2024–present).

Adjunct Faculty and Energy Economist — 2017–2022
Institut Teknologi Bandung, Bandung, Indonesia

- Delivered economic and financial analyses to Indonesia's Presidential Staff Office, SKK Migas (the national oil and gas regulator), Pertamina, and the West Java Governor's Office; appointed Governor's Expert Staff in Energy.
- Evaluated fixed-price versus incentive-based oil and gas contracts for SKK Migas (2021); analyzed a national fuel price equalization policy to inform subsidy review (2020); evaluated technoeconomic feasibility for geothermal carbon projects and oil field pilots.

Professional Service and Leadership

Journal referee: *Journal of Environmental Economics and Management* | **Founder:** Georgia Tech Economics Ph.D. Workshop (2025)
Committee member: School of Economics Chair Search Committee (2025–2026)

Honors and Awards

John Wylder Dissertation Fellowship (2026); EPICenter Summer Research Fellowship (2025); Georgia Tech Economics Third and Fourth Year Paper Prizes (2025, 2026); Berkeley/Sloan Diversity Fellowship (2025); Best Master Graduate, Institut Teknologi Bandung (2021).

Skills and Languages

Programming and tools: R, Stata, Python, SQL, Git, LaTeX.
Languages: English; Indonesian (native). **Work authorization:** F-1, STEM OPT eligible.